

FreeRTOS - Assignment 04

1. Implement an event-driven LED control system.

- Requirements
 - Create an aperiodic task to control LED1.
 - Configure a push button interrupt.
 - On button press:
 - ISR should notify the task.
 - Task should toggle LED1.
- Constraints
 - Do not poll the button.
 - Use:
 - Binary Semaphore OR
 - Task Notification

2. Two sensor tasks generate ADC readings.

Sensor Task 1
Sensor Task 2

Display resource is shared.

- Requirements
 - Tasks should not access UART directly.
 - Send formatted ADC values to queue.
 - Gatekeeper Task displays values on UART.